

Using AI Technology and Visualizations for Mathematical Exploration

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Global Transformation

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- ▶ Many tasks are being automated, while others are reducing their costs, difficulty, and execution time.
- ▶ The way we solve problems is changing.
- ▶ New jobs and industries will appear, others will disappear.
- ▶ Well being of people is always a priority, but it is hard to predict the consequences of changes. We need to choose the path to follow.

What is the role of universities, educational institutions, and us as teachers in the face of these changes?

AI is transforming the world

- ▶ In the short term, artificial intelligence provides us with tools and possibilities to make traditional education richer, more personalized, creative, and efficient.
- ▶ Along with these opportunities, we will also increasingly have the possibility of avoiding learning and leaving our tasks to machines. Is this good?
- ▶ The teacher's role is to provide students with the necessary skills and knowledge for their optimal performance in their respective roles in society. We must stay one step ahead.

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I Know Kung Fu - Matrix (1999) - Movie Clip HD Scene



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AI tools and possibilities in education

Educational optimization

AI offers education advanced tools for pedagogical innovation. Furthermore, it allows for the development of educational content that adapts to the individual needs of each student.

- ▶ Automated assessment systems with instant feedback.
- ▶ The creation of virtual classrooms, teachers, and interaction environments.
- ▶ Applications for identifying learning patterns and personalized tutoring systems.
- ▶ Translation of educational material and adaptation of content for specific social contexts.
- ▶ A diversity of creative examples! What else can you imagine?

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Work in progress...

Pedagogical Challenges of AI in Education



- ▶ Minimization of effort.

- ▶ Opportunities for fraud and presenting external content as one's own.
- ▶ Distractions that can be addictive.
- ▶ Proper handling of personal information and privacy.

How can we create interesting challenges for our students that allow them to learn and develop, where AI is a tool to be mastered rather than a magic box that solves everything?

Math Education: New opportunities

Common Issues:

- ▶ Boring, repetitive, pointless
- ▶ Too complicated
- ▶ Fear of the "Wrong Answer"

The New AI Reality:

- ▶ Typical math problems are solved in seconds
- ▶ AI can answer your questions and teach you
- ▶ AI can be your creative partner and help you visualize

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New priorities

- ▶ Understanding over memorizing
- ▶ Driven by curiosity: It must be fun!
- ▶ Problem solving: Challenges, get the feeling of success

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- ▶ Sometimes math examples are too abstract.
- ▶ Formulas are often too complicated.
- ▶ If you cannot handle it to do something and you cannot see it in action, what is the point?

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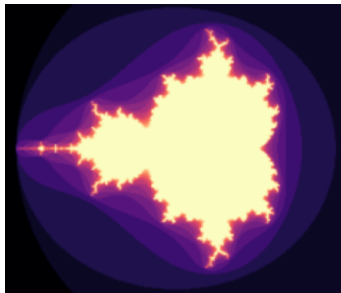
- ▶ Using **computers** we can make abstract math objects appear in front of us (as digital objects).
- ▶ They make easier the complicated computations.
- ▶ With code we can create, interact and visualize these objects.
- ▶ AI helps us to get the code very quickly.

If you are curious, you can explore endlessly!

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(I have plenty of examples, for this presentation I will briefly talk about two of them.)

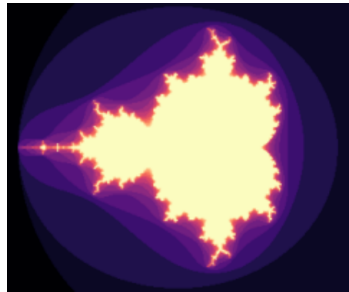
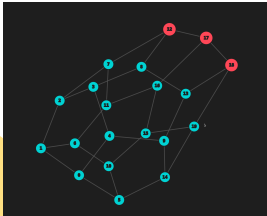
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- ▶ Fractals: [colab](#)
- ▶ Drawing graphs: [webpage](#)



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Text → Numbers → Math Formula → Answer → Text Answer

- The one question to answer them all: Which word is more likely to appear after this text? (find one word at a time)
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Without playing with math and computers these formulas would never have been found

It is more fun when we can **visualize** the results

- ▶ Math can be creative: nice combination of sums, multiplications, functions ($\sin$, $\cos$, e^x , ...), probabilities and calculus.
- ▶ We can explore among different formulas, find different results and applications.



The final remarks

- ▶ The Role of the Teacher: Guiding students through the noise to find high-quality learning material.
- ▶ How to engage the students so that they begin to follow their own path of exploration?
- ▶ We need to stay one step ahead in using technological tools to find the right challenges for students.

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